



INCOSE

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INCOSE Knowledge Systems WG

Context is King



Common Fundamentals



Ontology Mapping Exercise

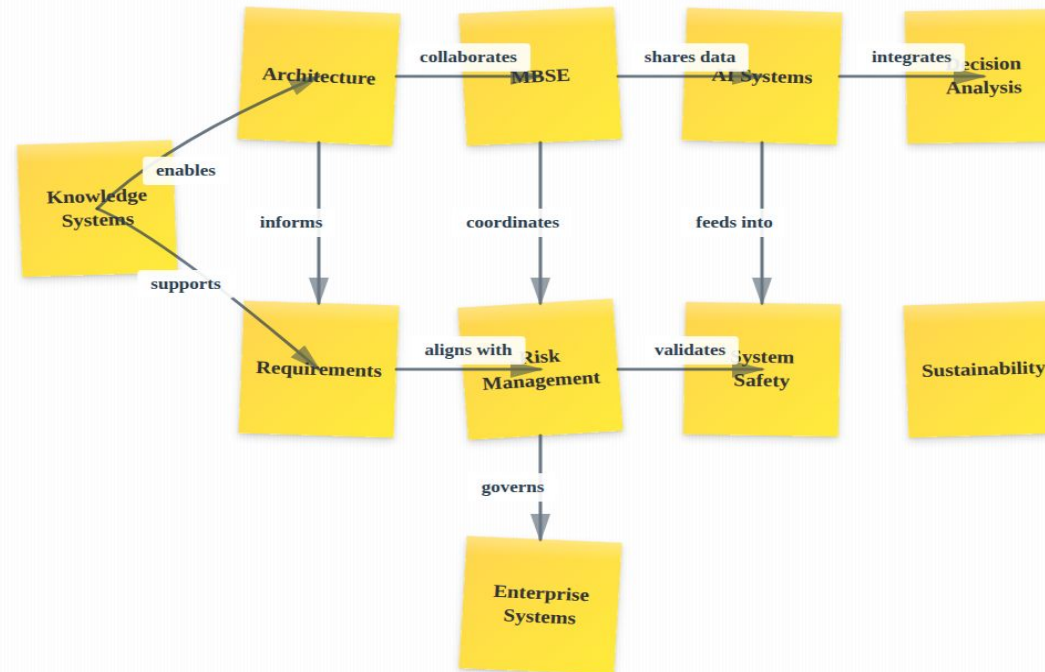
Working Groups Network

- 1 Identify Entities**
Place working groups on sticky notes (nodes in the network)
- 2 Draw Connections**
Connect groups that collaborate or interact (edges in the graph)
- 3 Label Relationships**
Describe the nature of each connection (relationship types)
- 4 Formalize Structure**
Transform the visual into a formal ontology model

Visual Key

-  Working Group (Node)
-  Relationship (Edge)

Network Diagram



Engineering

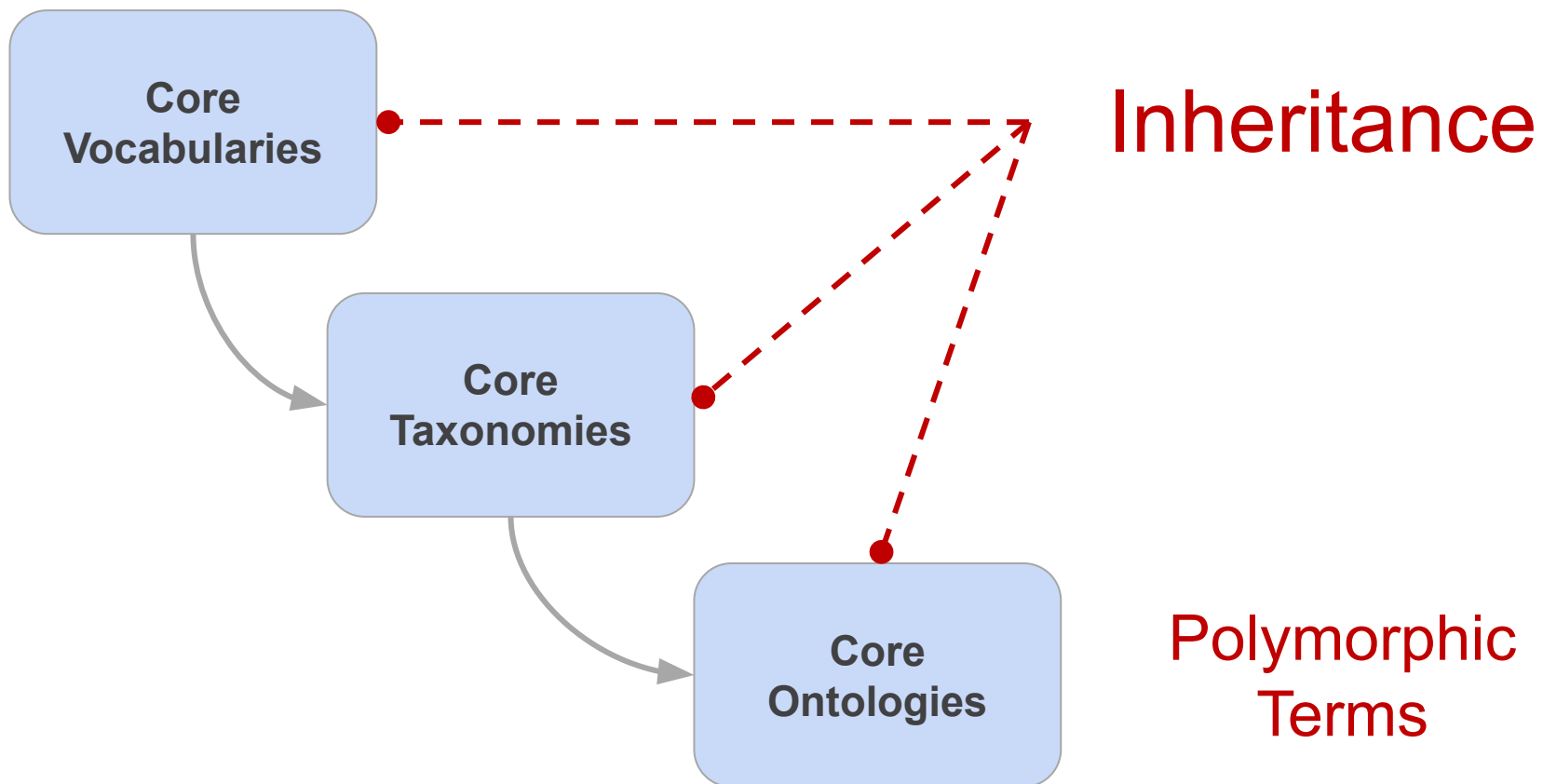
Systems

Policy

Domains



KSWG Owns the Core: Top-Down, Object Design



Engineering

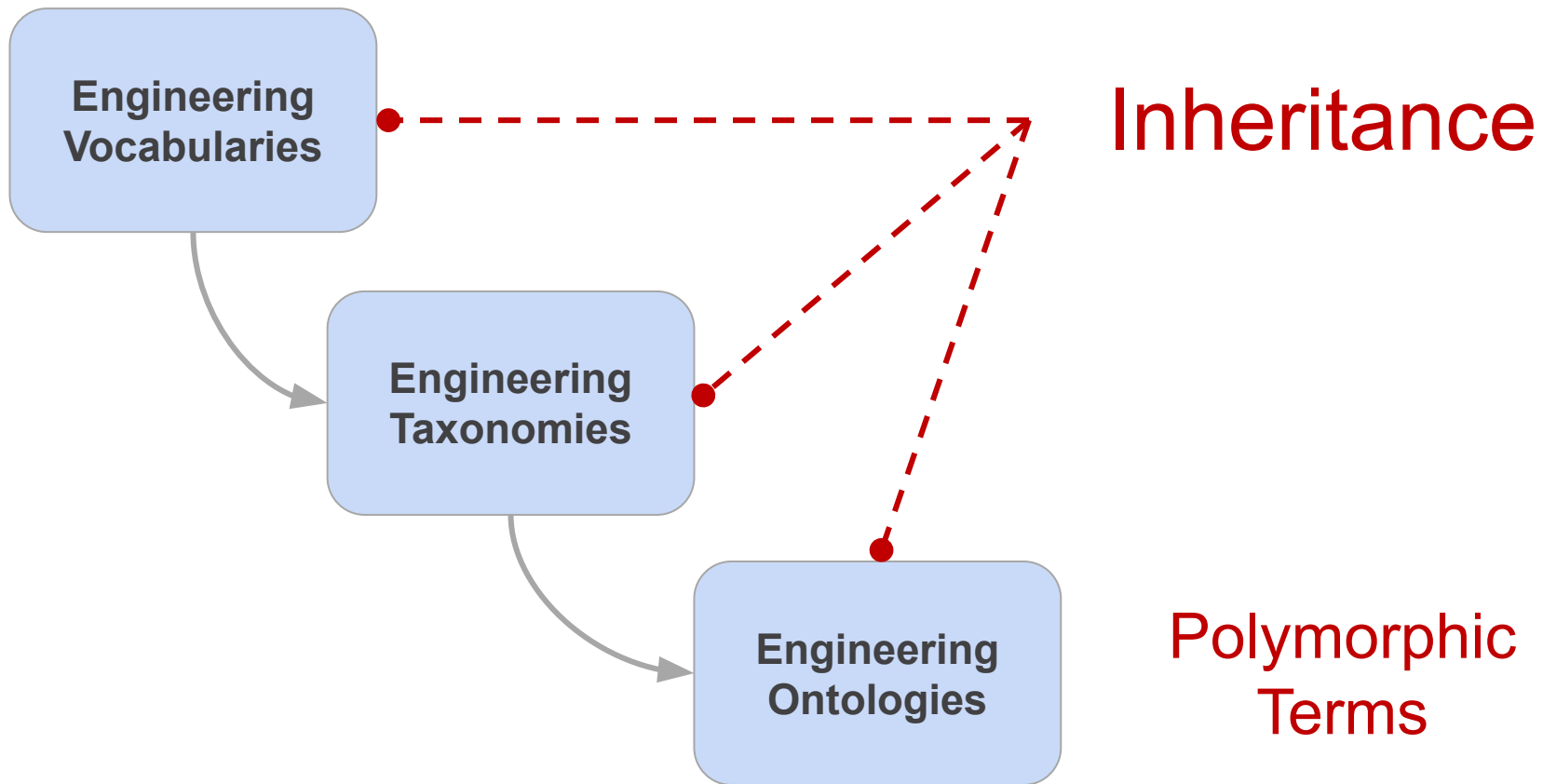
Systems

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Engineering Disciplines Own Their Disciplines



Engineering

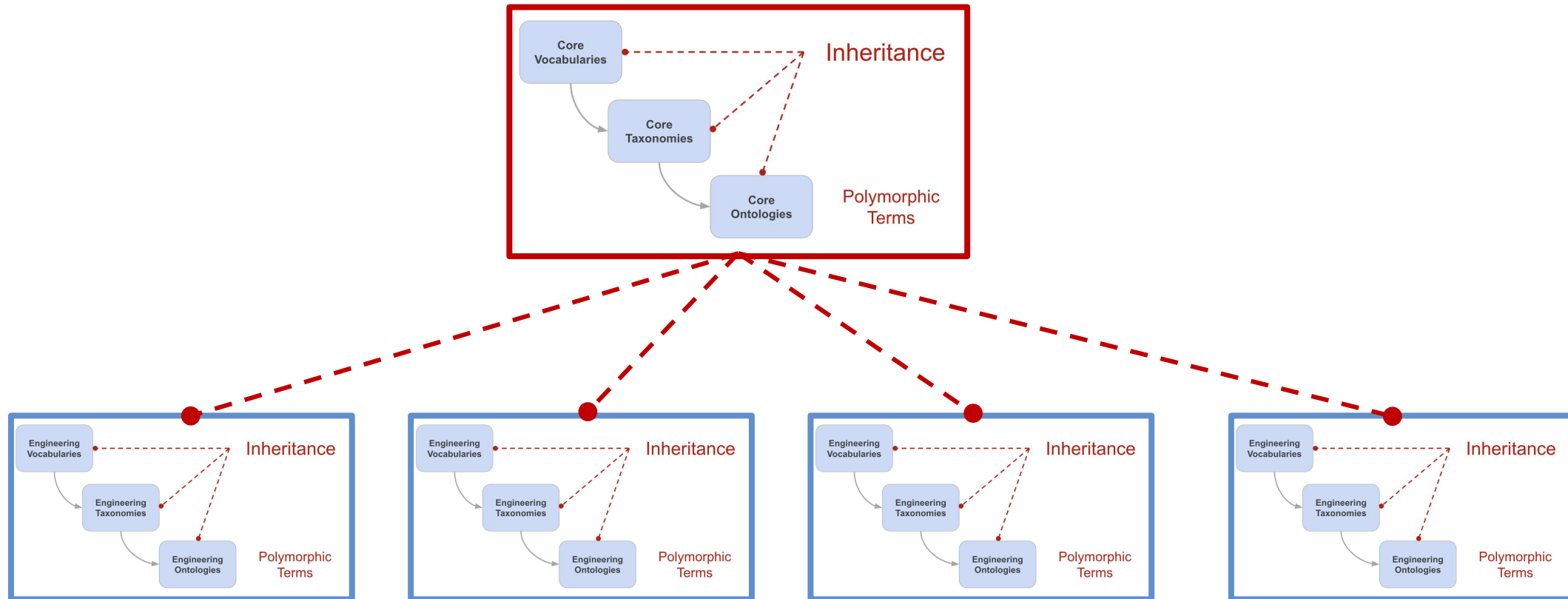
Systems

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Federated Topology of Engineering Knowledge



Engineering

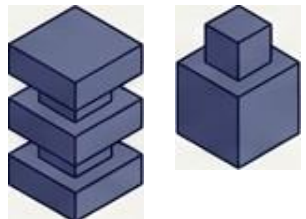
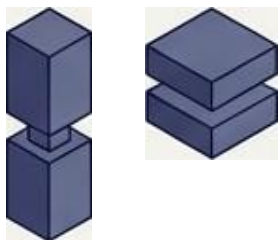
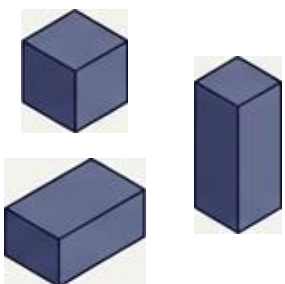
Systems

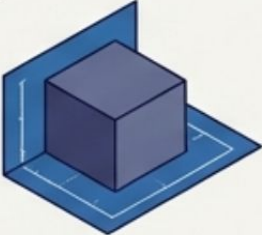
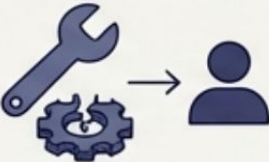
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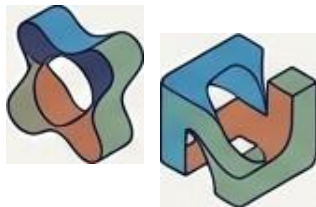
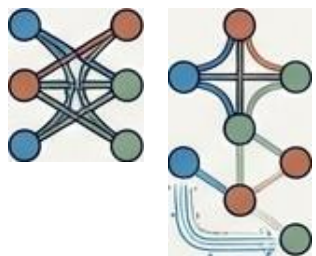
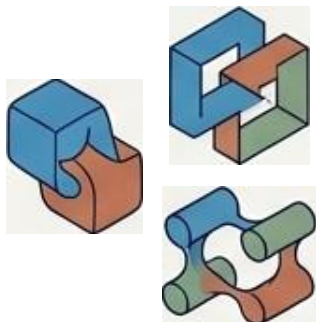
Domains



The transition from static designs to living systems



Static Engineering		Living Systems	
Fundamental State		Fundamental State	
			
A product is designed, built, and remains exactly as constructed.		Assets understand their own purpose, environment, and performance data.	
Response Mechanism		Response Mechanism	
			
Relies on human intervention for maintenance, patching, or adaptation.		Achieves systemic autonomy and continuous evolution without human input.	
Ultimate Goal		Ultimate Goal	
			
Meeting baseline specifications at the moment of deployment.		Maintaining integrity and function in unpredictable, complex environments.	



Engineering

Systems

Policy

Domains



Structuring the federated engineering ecosystem

Core Capability	Impact on the Lifecycle	Ultimate Outcome
Federated Interoperability	Enables design-time collaboration across disparate global teams.	Drastically reduced design cycles and elimination of re-work.
Autonomous Reasoning	Performs run-time safety checks and self-healing protocols.	Guaranteed high reliability in unpredictable Edge environments.
Knowledge Graphs	Ensures unbroken traceability from initial requirement to asset retirement.	Assured regulatory compliance and verifiable safety.

Engineering

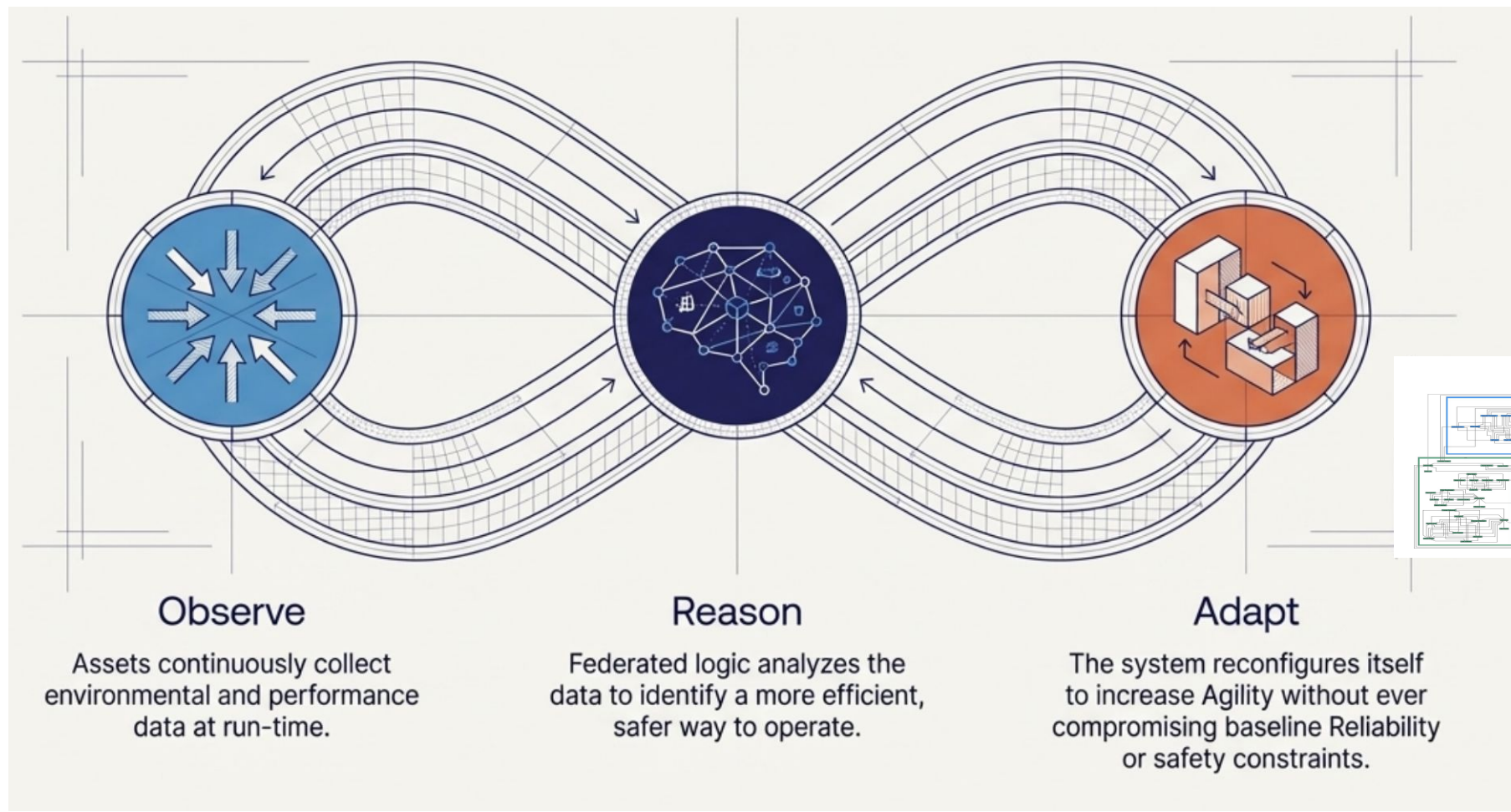
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The Engine of Autonomy: Self-Evolving Engineering



Engineering

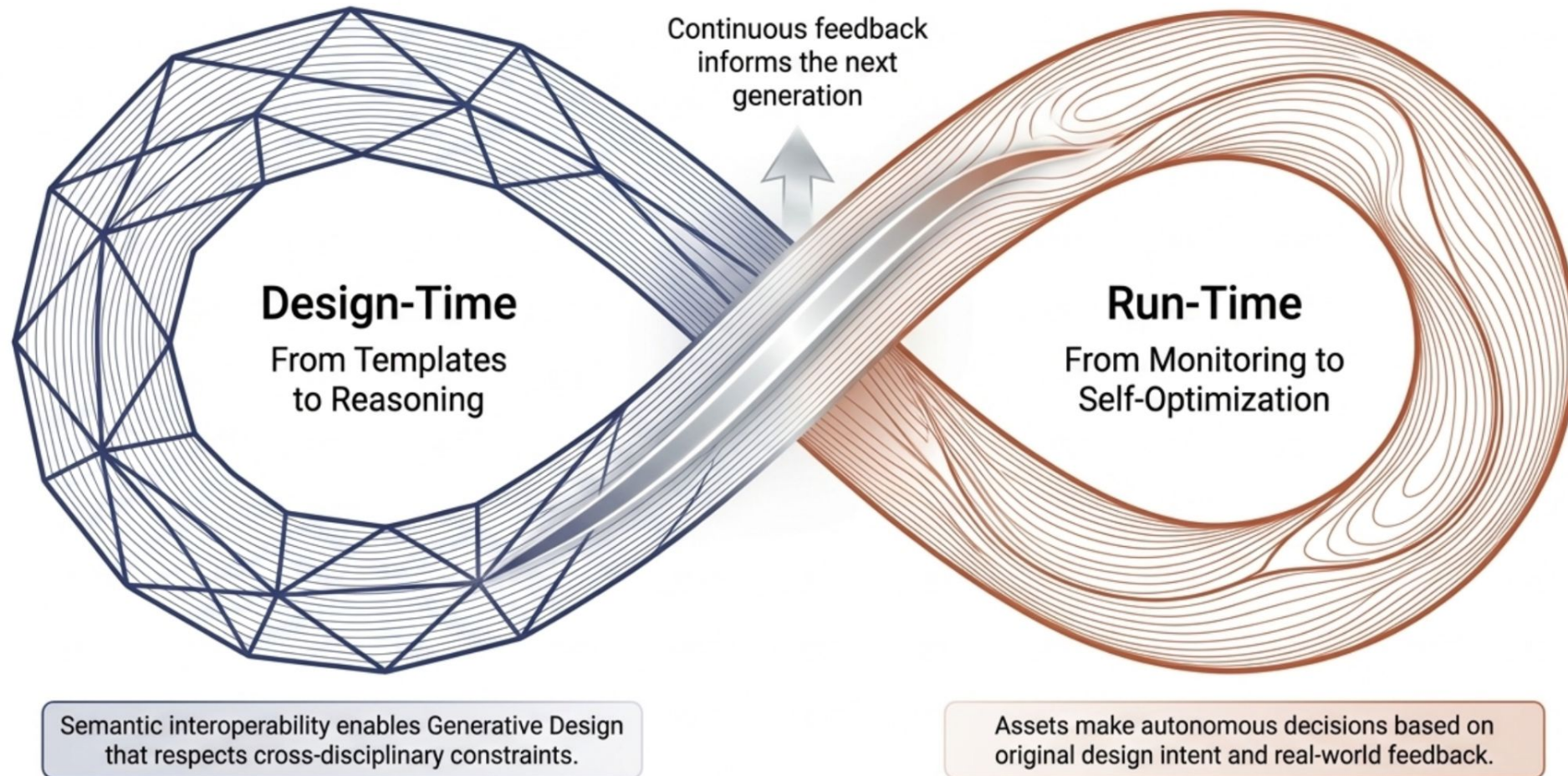
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Closing the self-evolving engineering loop



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